

Proji — структура БД (по PDF-ТЗ)

Источники (без отклонений от DDL в ТЗ):

- 1. Claud TZ DB structure.pdf — proj-i-data-layer (ядро, граф, канбан, auth)
- 2. proj-i_backend_detailed_specification.pdf — классическая REST-модель (12 таблиц)
- 3. PROJ-I_Backend_TZ_Extended.pdf — расширенный CORE (31 сущность)

СУБД: PostgreSQL 16 · PK: UUID · Время: TIMESTAMPTZ

Как связаны три PDF

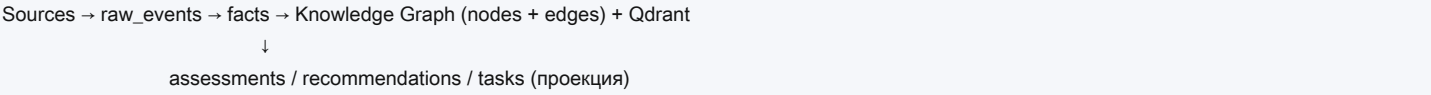
PDF	Роль	Когда использовать
Claud TZ	Data Layer + граф знаний + канбан-проекция	Бэкенд FastAPI, ingestion, LLM, Qdrant
Detailed spec	Классический таск-трекер + BI	NestJS/FastAPI REST, если без графа
Extended TZ	Enterprise-дорожная карта	V2 → Enterprise поверх CORE

Рекомендация для Proji: база = Claud TZ (обязательный DDL) + таблицы из Extended TZ, которых нет в Claud, но нужны фронту (conversations/messages для командного чата).

Полный SQL: schema.sql

ЧАСТЬ 1 — Claud TZ (Business Context Engine)

Архитектура (5 слоёв)



Принцип: raw_events и facts — append-only. Нет UPDATE/DELETE. Заккрытие факта/ребра — valid_to = now().

1.1. Ядро MVP — 6 таблиц (обязательный DDL)

raw_events — журнал сырых событий

```
CREATE TABLE raw_events (  
  id          BIGINT GENERATED ALWAYS AS IDENTITY,  
  org_id      UUID NOT NULL,  
  source      TEXT NOT NULL,          -- 'whatsapp' | 'telegram' | 'log' | 'doc' | ...  
  source_external_id TEXT,  
  occurred_at TIMESTAMPTZ NOT NULL,  
  ingested_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  actor_external_ref TEXT,  
  payload     JSONB NOT NULL,  
  content_hash TEXT NOT NULL,  
  PRIMARY KEY (org_id, id)  
);  
  
CREATE UNIQUE INDEX uq_raw_events_hash ON raw_events (org_id, content_hash);  
CREATE INDEX ix_raw_events_occurred ON raw_events (org_id, occurred_at);  
CREATE INDEX ix_raw_events_source ON raw_events (org_id, source, occurred_at);
```

outbox — Outbox Pattern (Celery)

```
CREATE TABLE outbox (  
  id          BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
  org_id      UUID NOT NULL,  
  event_id    BIGINT NOT NULL,  
  task_type   TEXT NOT NULL,          -- 'extract_facts' | 'embed_event'  
  status      TEXT NOT NULL DEFAULT 'pending', -- pending | processing | done | failed  
  attempts    INT NOT NULL DEFAULT 0,  
  last_error  TEXT,  
  created_at  TIMESTAMPTZ NOT NULL DEFAULT now(),  
  processed_at TIMESTAMPTZ  
);  
  
CREATE INDEX ix_outbox_pending ON outbox (status, created_at) WHERE status = 'pending';
```

facts — извлечённые LLM факты

```
CREATE TABLE facts (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  org_id      UUID NOT NULL,  
  fact_type   TEXT NOT NULL,          -- 'task_mentioned' | 'commitment' | 'blocker' | ...  
  subject_node_id UUID,  
  predicate   TEXT,  
  object_node_id UUID,  
  attributes  JSONB NOT NULL DEFAULT '{}',  
  source_event_id BIGINT NOT NULL,  
  extractor_version TEXT NOT NULL,    -- 'extract-v1'  
  confidence  NUMERIC(3,2) NOT NULL,  
  valid_from  TIMESTAMPTZ NOT NULL DEFAULT now(),  
  valid_to    TIMESTAMPTZ          -- NULL = актуален  
);  
  
CREATE INDEX ix_facts_subject ON facts (org_id, subject_node_id) WHERE valid_to IS NULL;  
CREATE INDEX ix_facts_source ON facts (org_id, source_event_id);  
CREATE INDEX ix_facts_type ON facts (org_id, fact_type, valid_from);
```

nodes — узлы графа знаний

```
CREATE TABLE nodes (
  id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id   UUID NOT NULL,
  node_type TEXT NOT NULL,      -- 'direction' | 'project' | 'sprint' | 'task' | 'person' | ...
  title    TEXT NOT NULL,
  props    JSONB NOT NULL DEFAULT '{}',
  status   TEXT,               -- 'auto_created' | ...
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE INDEX ix_nodes_type ON nodes (org_id, node_type);
CREATE INDEX ix_nodes_title ON nodes USING gin (to_tsvector('simple', title));
```

edges — связи графа

```
CREATE TABLE edges (
  id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id   UUID NOT NULL,
  from_node UUID NOT NULL REFERENCES nodes(id),
  to_node  UUID NOT NULL REFERENCES nodes(id),
  edge_type TEXT NOT NULL,      -- 'belongs_to' | 'assigned_to' | 'blocks' | ...
  props    JSONB NOT NULL DEFAULT '{}',
  source_fact_id UUID REFERENCES facts(id),
  valid_from TIMESTAMPTZ NOT NULL DEFAULT now(),
  valid_to   TIMESTAMPTZ
);

CREATE INDEX ix_edges_from ON edges (org_id, from_node, edge_type) WHERE valid_to IS NULL;
CREATE INDEX ix_edges_to ON edges (org_id, to_node, edge_type) WHERE valid_to IS NULL;
```

assessments + recommendations — заготовки аналитики

```
CREATE TABLE assessments (
  id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id   UUID NOT NULL,
  target_node_id UUID NOT NULL REFERENCES nodes(id),
  framework TEXT NOT NULL,
  result    JSONB NOT NULL,
  inputs_fact_ids UUID[] NOT NULL DEFAULT '{}',
  computed_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE TABLE recommendations (
  id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id   UUID NOT NULL,
  target_node_id UUID NOT NULL REFERENCES nodes(id),
  rec_type TEXT NOT NULL,
  rationale JSONB NOT NULL,
  status   TEXT NOT NULL DEFAULT 'proposed', -- proposed | accepted | rejected
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  resolved_at TIMESTAMPTZ
);
```

Multi-tenancy: у каждой таблицы обязателен org_id. Фильтр по org_id — во всех запросах и в Qdrant.

1.2. Auth и пользователи (Этап 7, Claud TZ)

users

```
CREATE TABLE users (
  id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email    TEXT NOT NULL UNIQUE,
  password_hash TEXT NOT NULL,
  full_name TEXT NOT NULL,
  is_active BOOLEAN NOT NULL DEFAULT true,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

memberships

```
CREATE TABLE memberships (
  user_id  UUID NOT NULL REFERENCES users(id),
  org_id   UUID NOT NULL,
  role     TEXT NOT NULL,          -- 'owner' | 'admin' | 'member' | 'viewer'
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  PRIMARY KEY (user_id, org_id)
);
```

refresh_tokens (коррекция JWT, раздел 20.1)

```
CREATE TABLE refresh_tokens (
  id        UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id   UUID NOT NULL REFERENCES users(id),
  token_hash TEXT NOT NULL UNIQUE,
  expires_at TIMESTAMPTZ NOT NULL,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  revoked_at TIMESTAMPTZ
);
```

RBAC: owner > admin > member > viewer. org_id только из membership, не из body запроса.

1.3. Канбан-проекция (Этап 8, Claud TZ)

Задача в графе = nodes.node_type = 'task'. Для доски — типизированная проекция:

```
CREATE TABLE tasks (
  node_id      UUID PRIMARY KEY REFERENCES nodes(id),
  org_id       UUID NOT NULL,
  project_node_id UUID NOT NULL REFERENCES nodes(id),
  column_status TEXT NOT NULL DEFAULT 'todo', -- todo | in_progress | review | done | ...
  order_index  NUMERIC(20,10) NOT NULL,
  assignee_node_id UUID REFERENCES nodes(id),
  due_at       TIMESTAMPTZ,
  rice_reach   NUMERIC(10,2),
  rice_impact  NUMERIC(10,2),
  rice_confidence NUMERIC(5,2),
  rice_effort  NUMERIC(10,2),
  updated_at   TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE INDEX ix_tasks_board ON tasks (org_id, project_node_id, column_status, order_index);
```

RICE: считается на лету: (reach × impact × confidence) / effort — не хранить итоговый score.

Batch: PATCH /v1/tasks/batch — до 100 элементов, атомарная транзакция.

Спринты (проекция):

```
CREATE TABLE sprints (
  node_id      UUID PRIMARY KEY REFERENCES nodes(id),
  org_id       UUID NOT NULL,
  project_node_id UUID NOT NULL REFERENCES nodes(id),
  name         TEXT NOT NULL,
  start_date   TIMESTAMPTZ,
  end_date     TIMESTAMPTZ,
  status       TEXT NOT NULL DEFAULT 'planned'
);
-- tasks.sprint_id → sprints.node_id (добавить колонку при необходимости)
```

1.4. Продуктовые таблицы (Claud TZ, этапы 9–10)

notifications

```
CREATE TABLE notifications (
  id        UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id    UUID NOT NULL,
  user_id   UUID NOT NULL REFERENCES users(id),
  notif_type TEXT NOT NULL,          -- mention | deadline | assignment
```

```
payload    JSONB NOT NULL,  
is_read    BOOLEAN NOT NULL DEFAULT false,  
created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

files (S3 Presigned)

```
CREATE TABLE files (  
  id        UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  org_id    UUID NOT NULL,  
  node_id   UUID REFERENCES nodes(id),  
  s3_key    TEXT NOT NULL,  
  filename  TEXT NOT NULL,  
  mime_type TEXT NOT NULL,  
  size_bytes BIGINT,  
  status    TEXT NOT NULL DEFAULT 'pending',  
  uploaded_by UUID REFERENCES users(id),  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

documents (wiki, отчёты)

```
CREATE TABLE documents (  
  id        UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  org_id    UUID NOT NULL,  
  node_id   UUID REFERENCES nodes(id),  
  title     TEXT NOT NULL,  
  content   TEXT,  
  doc_type  TEXT NOT NULL,  
  file_id   UUID REFERENCES files(id),  
  author_id UUID REFERENCES users(id),  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

chat_messages (AI-чат, SSE)

```
CREATE TABLE chat_messages (  
  id        UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  org_id    UUID NOT NULL,  
  user_id   UUID REFERENCES users(id),  
  context_node_id UUID REFERENCES nodes(id),  
  role      TEXT NOT NULL,      -- user | assistant | system  
  content   TEXT NOT NULL,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);  
CREATE INDEX ix_chat_messages_context ON chat_messages (org_id, context_node_id, created_at);
```

1.5. Qdrant (не PostgreSQL)

Коллекция proj_i_chunks: cosine, dim=768, модель intfloat/multilingual-e5-base.

Payload: org_id, source_event_id, chunk_index, node_ids[], occurred_at, source.

1.6. API-префиксы (Claud TZ, раздел 20.6)

Ter	Префикс	Содержимое
auth	/v1/auth/	login, refresh, logout
events	/v1/events	ingestion
graph	/v1/nodes, /v1/edges, /v1/graph/	граф
tasks	/v1/tasks/, /v1/projects/	канбан, batch, RICE
ai	/v1/ai/	chat SSE

search	/v1/search	семантика
files	/v1/files/	presign, confirm
documents	/v1/documents/	wiki
notifications	/v1/notifications/	список
ws	/ws	WebSocket

ЧАСТЬ 2 — Detailed Specification (12 таблиц)

Классическая модель без графа. Можно использовать как альтернативный слой или мигрировать в nodes/edges.

#	Таблица	Назначение
1	users	Пользователи (role: admin, manager, employ
2	business_lines	Бизнес-направления
3	projects	Проекты
4	project_members	Участники (PK: project_id + user_id)
5	sprints	Спринты
6	tasks	Задачи (status: backlog, todo, doing, in_r
7	task_dependencies	Зависимости задач
8	documents	База знаний
9	chat_messages	ИИ-чат (role: user, ai, system)
10	raw_activity_logs	Внешние логи (≈ raw_events в Claud TZ)
11	analysis_insights	ИИ-анализ (JSONB + GIN)
12	recommendations	Рекомендации (target_type: task, project,

Поля ключевых таблиц (как в PDF)

Поле	Тип
id	UUID PK
email	VARCHAR(255) UNIQUE NOT NULL
password_hash	VARCHAR(255) NOT NULL
full_name	VARCHAR(255) NOT NULL
role	ENUM admin, manager, employee, guest
avatar_url	VARCHAR(500)
created_at	TIMESTAMPTZ

Поле	Тип
id	UUID PK
project_id	UUID FK
sprint_id	UUID FK NULL
parent_id	UUID FK NULL
title	VARCHAR(255)
description	TEXT
status	ENUM backlog, todo, doing, in_review, done
priority	ENUM low, medium, high, critical
assignee_id, creator_id	UUID FK
order_index	FLOAT DEFAULT 0
rice_reach, rice_impact, rice_confidence,	NUMERIC
due_date, created_at, updated_at	TIMESTAMPTZ

```
CREATE VIEW view_tasks_with_rice AS
SELECT *, CASE WHEN rice_effort > 0
  THEN (rice_reach * rice_impact * rice_confidence) / rice_effort END AS rice_score
FROM tasks;
```

Поле	Тип
------	-----

id	UUID PK
source	VARCHAR(50) — slack, telegram, github, jir
project_id	UUID FK NULL
user_id	UUID FK NULL
payload	JSONB NOT NULL
created_at	TIMESTAMPTZ

GIN index на payload.

ЧАСТЬ 3 — Extended TZ (31 сущность PROJI CORE)

#	Таблица	Фаза	Соответствие
1	organizations	MVP	org_id в Claud TZ (отдельная таблица — до
2	organization_members	MVP	= memberships
3	business_directions	MVP	= nodes type=direction
4	teams	MVP	узлы + edges
5	projects	MVP	= nodes type=project + проекция
6	sprints	MVP	sprints проекция
7	tasks	MVP	tasks проекция
8	task_dependencies	V2	edges edge_type=blocks
9	task_comments	MVP	новая таблица (фронт)
10	documents	MVP	documents
11	conversations	MVP	для /messages (Zulip)
12	messages	MVP	командный чат
13	meetings	Enterprise	—
14	decisions	Enterprise	Decision Intelligence
15	risks	Enterprise	—
16	insights	V2	≈ assessments
17	recommendations	V2	recommendations
18	data_sources	V2	коннекторы
19	source_items	V2	≈ raw_events
20	knowledge_chunks	V2	Qdrant + PG метаданные
21	embeddings	V2	Qdrant
22	frameworks	V2	SWOT, scrum_health...
23	framework_analysis	V2	≈ assessments
24	entity_relations	Enterprise	= edges
25	metrics	V2	Velocity, Lead Time...
26	metric_values	V2	временные ряды
27	ai_agents	Enterprise	AI Agent Registry
28	agent_memory	Enterprise	Decision Memory
29	event_log	Enterprise	шина событий
30	notifications	MVP	notifications
31	audit_log	MVP	аудит

Дополнение под фронт (Extended п.11–12, нет в Claud DDL)

```
CREATE TABLE organizations (  
  id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  name    TEXT NOT NULL,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);  
  
CREATE TABLE conversations (  
  id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  org_id  UUID NOT NULL REFERENCES organizations(id),  
  stream  TEXT NOT NULL DEFAULT 'general', -- general | tasks | clients  
  type    TEXT NOT NULL,                  -- team_general | task | external  
  task_node_id  UUID REFERENCES nodes(id),  
  title   TEXT,
```

```
last_message_at TIMESTAMPTZ,  
created_at    TIMESTAMPTZ NOT NULL DEFAULT now()  
);  
  
CREATE TABLE messages (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  org_id      UUID NOT NULL,  
  conversation_id UUID NOT NULL REFERENCES conversations(id),  
  author_id   UUID REFERENCES users(id),  
  body        TEXT NOT NULL,  
  metadata    JSONB NOT NULL DEFAULT '{}',  
  created_at  TIMESTAMPTZ NOT NULL DEFAULT now()  
);  
CREATE INDEX ix_messages_conv ON messages (conversation_id, created_at);
```

Итоговый список таблиц PostgreSQL (рекомендуемый набор)

MVP (запуск фронта + data layer)

- organizations
- users
- memberships
- refresh_tokens
- raw_events
- outbox
- facts
- nodes
- edges
- tasks (проекция канбан)
- sprints (проекция)
- conversations
- messages
- chat_messages (AI)
- documents
- files
- notifications
- assessments
- recommendations

V2

- data_sources · 21. task_comments · 22. metrics · 23. metric_values · 24. frameworks · 25. audit_log

Enterprise

- ai_agents · 27. agent_skills · 28. agent_runs · 29. agent_memory · 30. meetings · 31. decisions · 32. decision_outcomes · 33. risks · 34. event_log · 35. feature_flags

Маппинг фронт → PDF-таблицы

Фронт	PDF-источник	Таблица
Канбан /tasks	Claud TZ §12	tasks + nodes
Сообщения /messages	Extended §11–12	conversations + messages
AI-панель	Claud §20.2 + Detailed §9	chat_messages
Команда /team	Claud nodes person + tasks.assignee_node_i	nodes, tasks
Документы	Claud §20.3	documents, files
Wazzup	Claud raw_events source=whatsapp	raw_events → facts → nodes
Onboarding org	Extended §1	organizations, memberships

Индексы (из PDF DoD)

```
CREATE INDEX ix_raw_events_payload ON raw_events USING GIN (payload); -- если нужен поиск в payload
CREATE INDEX ix_assessments_result ON assessments USING GIN (result);
-- B-Tree на все FK
-- GIN на JSONB: payload, analysis_data, rationale
```

DoD: batch 100 задач < 150ms · ingestion < 200ms · миграции только через Alembic/Prisma.